

Modeling safety reliability and unreliability in a mean-excess network equilibrium framework: Heterogeneous users with safety-conscious route choice sets

Umer Mansoor^{a,b}, Zhengyang Li^a, Anthony Chen^{a,b,*}

^a Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University, Kowloon, Hong Kong

^b Research Institute for Sustainable Urban Development, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong

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ABSTRACT

Empirical studies highlight the critical role of safety in route choice decisions and reveal the presence of traveler classes with heterogeneous safety preferences in a transportation network. However, safety has been largely overlooked in network equilibrium models, which have traditionally prioritized efficiency. This study proposes an integrated network equilibrium framework that models safety-conscious class-specific route choice sets while simultaneously incorporating travelers' concerns about the reliability and unreliability of safety and travel time. The route choice set for each class is defined based on their safety score requirements, with the score of each route quantified by its roadway characteristics. The reliability and unreliability aspects of safety variability for heterogeneous travelers are modeled using the α -reliable mean-excess concept, and their safety variability concerns are represented by a crash risk cost distribution. The proposed model ensures the reliability aspect of safe arrival at a specified confidence level " α " while also accounting for the unreliability aspect of encountering the extreme safety conditions, such as excessively large crash risk costs (high crash severity). Formulated as a variational inequality problem, the proposed model is solved using a two-stage algorithm. The first stage determines the route choice set for each class, while the second stage solves the network equilibrium model. Numerical experiments on both small-scale and real-world networks demonstrate how safety preferences influence route choices and subsequent flow patterns, showing that high safety-conscious travelers prioritize safer routes. The proposed framework offers transportation planners a way to accommodate the effect of reshaping travel behavior on demand patterns.

1. Introduction

1.1. Background and motivation

Transportation network safety is a critical concern, driving the need for safety-conscious planning, the process of embedding safety considerations into transportation planning to enhance overall system effectiveness (Cai et al., 2019). Traditionally, planners have relied on macro-level crash prediction models, which are based on historical crash data, to guide decisions about zonal development and achieve specific safety targets (Abdel-Aty et al., 2013; Li et al., 2013). While these models have proven useful, they fail to account

* Corresponding author at: Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University, Kowloon, Hong Kong.
E-mail address: anthony.chen@polyu.edu.hk (A. Chen).

for travelers' safety preferences and behaviors. This limitation prevents planners from fully understanding how individual route choices, shaped by safety concerns, influence traffic flow patterns and, ultimately, network-wide safety. Bridging this gap is crucial for planning and designing transportation networks that not only meet aggregate safety goals but also align with real-world user behavior, allowing planners to proactively address safety issues.

This consideration is especially crucial in traffic assignment, a key step in transportation planning, where different route choice criteria have been used to assign traffic on road networks, such as minimum travel time routes (Bellman, 1958; Gendreau et al., 2015), environment-friendly routes (Xu et al., 2015), and the most reliable routes (Xu et al., 2015). Empirical studies have also highlighted the importance of travel safety in travelers' route choice decision-making (Abdel-Aty et al., 1995; Amirgholy et al., 2017; Yannis et al., 2005). This is supported by both revealed-preference data, showing that observable route attributes like intersection density influence choices (Hao et al., 2025), and stated-preference studies, which confirm that many travelers are willing to accept longer trips for improved safety (Rizzi and Ortúzar, 2003; Yannis et al., 2005), demonstrating heterogeneous safety preferences (Kavta et al., 2025).

Crucially, travelers do not make these decisions by calculating low-probability crash statistics. Instead, they assess safety through observable proxies: static attributes (e.g., road class, intersection density) signal inherent risk (Dijkstra et al., 2007), while traffic conditions (e.g., speed, congestion) signal immediate risk levels (Elvik et al., 2004). This assessment process, well-documented in risk-perception literature (Nordfjærn & Rundmo, 2009; Rundmo & Nordfjærn, 2013, 2017), explains why safety perceptions influence behavior even when objective crash risks are small. This cognitive process means that safety assessment is not confined to extreme conditions but operates continuously during routine travel. Even in normal commuting, travelers constantly evaluate routes using these observable proxies, forming persistent safety perceptions that influence their choices. Consequently, safety functions as a routinely present factor in route selection, with its relative importance varying across individuals based on their heterogeneous safety consciousness (Kavta et al., 2025).

Despite this behavioral evidence, safety, specifically crash risk, has received relatively little attention in traffic assignment models (Han et al., 2023; Huang et al., 2020; Ma & Zhang, 2019). As we move into an era of intelligent and connected vehicles, the availability of real-time trip-related safety information highlights the need to incorporate safety considerations into traffic assignment models (Carmody & Sowers, 2019; Ghoul et al., 2023; Hoseinzadeh et al., 2020). Integrating safety into traffic assignment enhances safety-conscious planning by offering a behaviorally informed framework that explicitly incorporates travelers' safety behaviors and preferences as a core element of transportation planning.

Operationally, safety can be expressed as the crash risk potential of a vehicle traversing a road network (Chandra, 2014). This risk is influenced by various factors including roadway characteristics, environmental conditions (e.g., weather), human behavior, and traffic conditions. Roadway characteristics such as road classification, intersection density, and geometric design are fixed and inherent to a specific location. In contrast, factors like weather, human behavior and traffic conditions are subject to change over time, inducing variability in the crash risk of a road site, which fluctuates around a mean crash risk value. This mean crash risk value represents the average safety condition of the road (Han et al., 2023; Huang et al., 2020) while variability arises from uncertainties associated with changing traffic conditions, volatility in driving behavior (Wali et al., 2018; Wang et al., 2015), and environmental changes. Consequently, predicting the certainty of completing a trip safely is challenging, and a single deterministic value for trip safety may not be realistic. Although travelers may not know the exact crash risk of a route, they may be familiar with the route's crash risk variability based on their daily traveling experiences. For example, travelers may have knowledge of routes that have a higher probability of severe crashes, such as those with more truck volume or high-speed vehicles (Kadeha et al., 2021; Kidando et al., 2019; Uddin & Huynh, 2020). Therefore, to ensure trip safety and avoid routes with higher probabilities of severe or fatal crashes (i.e., extremely high crash costs), travelers are expected to consider the safety variability of a route in their habitual route choice process.

Moreover, empirical studies indicate that travelers on the road exhibit varying degrees of safety-consciousness, leading to heterogeneous safety preferences (May et al., 2008). For instance, young drivers often engage in higher speeds, underestimate potential crash risks, and overestimate their driving skills (Castellà & Pérez, 2004; Deery, 1999; Machin & Sankey, 2008; McKenna & Horswill, 2006). In contrast, older drivers tend to be more defensive and risk-averse (Simons-Morton et al., 2014). Incorporating these heterogeneous safety preferences into traffic assignment models is essential, particularly when addressing safety variability. Additionally, travelers with higher safety preferences may not consider all available routes in their choice set, instead excluding those with higher predefined safety risks. These exclusions are often based on fixed roadway characteristics such as road functional classification (e.g., highway or local road), intersection density, and total travel distance (Dijkstra, 2013; Dijkstra et al., 2007). Consequently, each traveler class may have a distinct route choice set aligned with its safety preferences (Li & Huang, 2019). Therefore, this study aims to examine the effect of travelers' heterogeneous safety preferences on route choices and equilibrium flow patterns within a transportation network, from two perspectives: (a) travelers' safety-conscious route choice sets, and (b) heterogeneous safety reliability preferences.

1.2. Related studies

A growing body of research has begun to incorporate travelers' safety concerns into traffic assignment models. For example, Ma and Zhang (2019) modeled expected safety using crash risk as a proxy, quantified via Safety Performance Functions (SPFs) for highways and freeways. Their findings indicated that link flow patterns from safety-oriented route choices differ from those driven solely by travel time minimization. However, this study did not simultaneously account for travelers' joint valuation of time and safety. Similarly, Sharifi and Shabaniverki (2016) incorporated safety by adding mean crash delay to the generalized link cost, modeling crash risk as a function of opposing traffic volumes under symmetric link interactions. Zhu et al. (2009) proposed a reliability-based stochastic user equilibrium model to explore the effect of turn delay uncertainties on the route choice behavior at signalized intersections.

The proposed model was applied to explore the optimal cycle times for crash reduction and effective travel time reduction at signalized intersections.

Furthermore, Huang et al. (2020) extended expected safety cost models by considering the safety variability (reliability) in the route choice for users with heterogeneous crash risk aversions (low and high-risk-averse users). They employed a Crash Risk Cost (CRC) distribution and proposed the concept of *effective CRC*, defined as the mean CRC plus a safety margin, inspired by the Travel Time Budget (TTB) model (Lo et al., 2006; Shao et al., 2006; Xu et al., 2018). While TTB ensures on-time arrival at a given confidence level α , effective CRC analogously ensures safe arrival. However, Chen and Zhou (2010) noted that considering only TTB may inadequately capture travel time risk, as it overlooks the consequences of extremely long trips (the *unreliability* aspect). They therefore proposed the α -reliable mean-excess traffic equilibrium (METE) model, which accounts for both acceptable travel times (reliability) and excessively long trips (unreliability).

In the same vein, relying solely on *effective CRC* may not fully represent travelers' safety-risk preferences. Decision-making literature emphasizes that potential losses in worst-case scenarios significantly influence choices (Yamai & Yoshida, 2002). For safety-conscious travelers, avoiding routes with high probabilities of severe outcomes (fatal or serious injury crashes), such as crashes involving heavy vehicles (Anderson & Hernandez, 2017; Behnood & Mannering, 2019) or on undivided roads (Kadeha et al., 2021; Kidando et al., 2019), is critical. Thus, incorporating both the *reliability* and *unreliability* of safety variability into equilibrium models offers a more behaviorally realistic approach. Han et al. (2023) advanced this by integrating both aspects of safety variability into a speed-route joint choice model. Yet, their model does not incorporate travel time variability, which also shapes route choice. A comprehensive model should therefore account for variability in *both* safety and travel time.

Additionally, existing safety-inclusive assignment models typically assume all routes are available to every traveler (Han et al., 2023; Huang et al., 2020). In reality, more safety-conscious travelers may preemptively exclude high-risk routes based on fixed roadway attributes like road class, intersection density, or length (Dijkstra, 2013; Dijkstra et al., 2007). Li and Huang (2019) introduced heterogeneous route choice sets for users with different safety preferences using an exogenous safety index—a meaningful step, though one that does not derive safety levels directly from roadway characteristics. The concept of class-specific choice sets has also been explored in other contexts, such as by Lin and Leong (2014), though not with a safety focus. The preceding review highlights several interrelated limitations in existing safety-inclusive traffic assignment models, which are systematically addressed in the following section.

1.3. Research gaps

The above discussion identifies three interconnected gaps in the existing research (Section 1.2).

- Limited scope of safety modeling:** While studies have incorporated safety into equilibrium models, they primarily rely on traffic conditions (e.g., speed, flow) as risk determinants (Han et al., 2023; Huang et al., 2020; Ma & Zhang, 2019), neglecting the influence of roadway characteristics. Factors such as road functional class, route distance, and intersection density are essential for evaluating safety but remain underexplored, despite their empirical relevance in risk perception and route choice (Dijkstra, 2013; Dijkstra et al., 2007).
- Isolated treatment of dual uncertainties:** Second, although safety and travel time variability concerns have been studied individually, e.g., through the mean-excess travel time metric for travel time variability (Chen & Zhou, 2010; Chen et al., 2011; Xu

Table 1

Selected studies on safety, safety variability, travel time variability, and safety-based class-specific route choice sets in network equilibrium models.

Study	Safety cost	Time cost	Safety variability	Time variability	Class-specific choice set	Key limitations addressed by our work
Chen and Zhou (2010); Chen et al. (2011); Xu et al. (2018); Xu et al. (2014)	×	METT	×	✓	×	o Models only travel time reliability and unreliability with full choice sets.
Huang et al. (2020),	ECRC	ETT	✓	×	×	o Models only safety reliability, with full choice sets.
Han et al. (2023)	MECRC	ETT	✓	×	×	o Safety based on traffic conditions. o Models only safety reliability and unreliability with full choice sets.
Li and Huang (2019)	Predefined safety index	ETT	×	×	✓	o Safety based on traffic conditions o Class-specific choice sets o Exogenous safety index.
This study	Safety score & MECRC	METT	✓	✓	✓	Unified framework to: o <i>Simultaneously model safety & travel time reliability and unreliability</i> o <i>Model safety-based class-specific route choice sets.</i> o <i>Safety based on traffic conditions and roadway characteristics</i>

Note: ECRC: effective crash risk cost; MECRC: mean excess crash risk cost; ETT: expected travel time; METT: mean excess travel time.

et al., 2018; Xu et al., 2014) and the effective crash risk cost or mean-excess crash risk cost for safety variability (Han et al., 2023; Huang et al., 2020), their simultaneous consideration in network equilibrium models has been largely overlooked. This integration is crucial for capturing holistic traveler behaviors under dual uncertainties.

3. **Overlooked heterogeneity in route consideration:** Existing models assume all travelers consider the same universal set of routes. This fails to capture the plausible behavior where safety-conscious users pre-filter their choice sets, avoiding routes perceived as high-risk. Preliminary work by Li and Huang (2019) introduced perceived choice sets but used an exogenous safety index, highlighting the need for an integrated approach.

Collectively, these gaps point to the need for a network equilibrium framework that simultaneously accounts for safety quantified via roadway attributes, dual uncertainties, and safety-based choice set heterogeneity. This study is designed to address these gaps, as summarized in Table 1.

1.4. Study contributions

To address the above research gaps, this study develops a comprehensive network equilibrium framework that integrates roadway characteristics and traffic conditions to model travelers' safety concerns. The key contributions of this study are as follows:

1. **Modeling safety-conscious route choice sets:** The proposed framework models and analyzes the effect of safety-based class-specific route choice sets on equilibrium flow patterns, extending prior assumptions of complete choice sets (Han et al., 2023; Huang et al., 2020). This accounts for heterogeneous travelers with varying degrees of safety consciousness.

Table 2
Notations.

Symbol	Description
Sets:	
$G(N,A)$	a road network composed of nodes and links
N	set of nodes
A	set of links, indexed by a
P	Set of routes, indexed by p
W	set of origin destination pairs, indexed by w
M	Set of traveler classes, indexed by m
Parameters/inputs:	
$\Gamma/\bar{\Gamma}$	adjustment coefficients
$\omega/\bar{\omega}$	parameters of travelers' risk perceptions
t_a^0	free-flow travel time of link a
c_a	capacity of link a
q_w^m	travel demand between an O-D pair w for user class m
Variables	
$f_{p,m}^w$	traffic flow on route p between O-D pair w for traveler class m
x_a	flow of link a
Others	
t_a	mean travel time of link a
r_a	CRC distribution of link a
μ_a	mean CRC per unit exposure of link a
σ_a	standard deviation of CRC per unit exposure of link a
$ls_{lr}^{p,w}$	length share of local road on route p between O-D pair w
$ls_{cr}^{p,w}$	length share of collector road on route p between O-D pair w
l_p^w	total length of route p between O-D pair w
$id_{p,w}$	intersection density on route p between O-D pair w
$\gamma_c^{p,w}$	score of a criterion c for route p between O-D pair w
γ_c	weight factor for criterion c
US_p^w	unsafety score for route p between O-D pair w
Θ_p^w	safety score of route p between O-D pair w
Θ_{req}^m	safety score requirement of traveler class m
r_p^w	CRC distribution of route p between O-D pair w
δ_a^p	route-link incidence parameter
μ_p^w	mean CRC of route p between O-D pair w
σ_p^w	standard deviation of CRC of route p between O-D pair w
$R_{p,m}^w$	effective CRC of route p between O-D pair w for traveler class m
$G_{p,m}^w$	generalized travel cost of route p between O-D pair w for class m
T_p^w	mean travel time of route p between O-D pair w for traveler class m
σ_p^w	standard deviation of travel time of route p between O-D pair w
$\psi_{p,m}^w$	mean excess CRC of route p between O-D pair w for traveler class m
η_p^w	mean excess travel time of route p between O-D pair w
Λ	Cost converting factor between safety and travel time

2. **Modeling of dual uncertainties:** The framework incorporates a traffic equilibrium model that simultaneously accounts for travelers' concerns related to both the reliability and unreliability of safety and travel time variabilities, considering traveler classes with heterogeneous safety reliability preferences. This integration builds on prior models of individual variability, including travel time variability (Chen & Zhou, 2010; Chen et al., 2011; Xu et al., 2018) and safety variability (Han et al., 2023; Huang et al., 2020), to provide a unified approach that captures holistic traveler behavior under dual uncertainties.
3. **Framework for proactive safety planning:** The proposed framework enables planners to explicitly incorporate travelers' safety preferences into the transportation planning process, allowing for the development of targeted safety improvement policies that are informed by behavioral insights.

In this study, the route choice set for each traveler class is determined based on their specific safety score requirements. This safety score is quantified using roadway characteristics such as road classification, intersection density and travel distance of each route (Dijkstra, 2013). As a result, each traveler class has a unique route choice set tailored to its specific safety requirements. Furthermore, the reliability and unreliability aspects of safety and travel time variabilities are modeled using the α -reliable mean-excess traffic equilibrium (METE) model (Chen & Zhou, 2010). Two traveler classes with heterogeneous safety preferences are considered in the METE model. These objectives aim to provide a deeper understanding of travelers' route choice behavior under heterogeneous safety concerns, contributing to a more comprehensive understanding of safety-conscious traffic assignment.

The remainder of the paper is organized as follows: Section 2 introduces the methodology for calculating the safety score of a route and explains the concepts of crash risk cost distribution and mean-excess crash risk cost (MECRC). It also formulates the mean-excess equilibrium model as a variational inequality (VI) problem. Section 3 presents the solution algorithm. Section 4 presents the results obtained from the model for two networks, while Section 5 discusses the practical implications of these findings. Finally, conclusions and future work are given in Section 6.

2. Methodology

2.1. Notations

All the notations used in the paper are listed in Table 2, unless otherwise specified.

2.2. Study framework

The framework of this study is illustrated in Fig. 1. It follows a *two-stage cognitive process* that captures how travelers integrate safety into route choice:

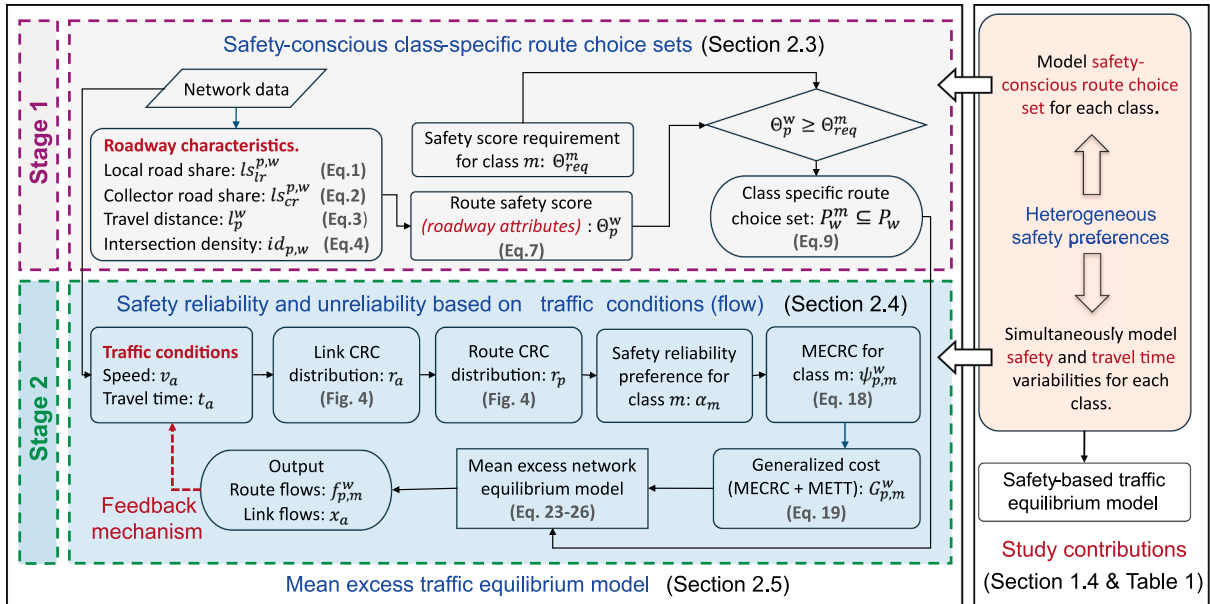


Fig. 1. Illustration of the proposed framework. Each box in the framework includes a Section or Equation number, indicating its specific location in the paper. The contributions outlined in Table 1 and Section 1.4 are emphasized in the rightmost box.

- **Stage 1 – Choice set formation:** Travelers first form a personalized route choice set based on inherent safety features (e.g., road type, intersection density). This stage uses route safety scores (Θ_p) to filter the network, determining which routes a traveler would even consider. This process is detailed in [Section 2.3](#).
- **Stage 2 – Mean excess traffic equilibrium model:** Within their individual choice sets, travelers evaluate routes by balancing travel time against the flow-dependent safety cost, modeled as a crash risk cost (CRC) distribution. Their sensitivity to the reliability and unreliability of this safety variability is formalized in [Section 2.4](#) using the mean excess crash risk cost of a route (ψ_p). The CRC for a route depends on traffic speed, which is itself a function of the assigned flow. This interdependence creates a critical feedback mechanism: assigned flow affects speed, which updates the CRC, which in turn re-weights route costs and leads to reassignment of flow.

Finally, these safety preferences and the flow-risk feedback loop are embedded within a mean-excess traffic equilibrium model. This model, presented in [Section 2.5](#), solves for the equilibrium state where no traveler can reduce their generalized cost (METT + MECRC).

2.3. Safety level of a route based on roadway characteristics

Routes with varying roadway characteristics exhibit different levels of safety. Specifically, routes differ in terms of their functional classes, intersection density, and total travel distance, all of which influence travel safety. Below, we explain the procedure for computing the safety score of a route based on several criteria related to its fixed characteristics ([Dijkstra, 2013; Dijkstra et al., 2007](#)).

2.3.1. Definition of different criteria

(i). Minimize the share of local roads on a route

The major function of local roads is to provide direct access to residential areas, businesses and other local areas. These roads are not meant for through traffic. A higher proportion of local roads on a route may lead to more transitions from high-speed facilities to low-speed roads ([Pervaz et al., 2024](#)). Therefore, the length of this road category on a route should be minimal. The percentage length share ($ls_{lr}^{p,w}$) of local roads on a route is calculated as the length of local roads ($l_{lr}^{p,w}$) to the total length (l_p^w) of a route p between OD pair w ([Eq. 1](#)) ([Dijkstra, 2013](#)).

$$ls_{lr}^{p,w} = \frac{l_{lr}^{p,w}}{l_p^w} \times 100, \forall p \in P_w, w \in W \quad (1)$$

(ii). Minimize the share of collector roads on a route

The collector roads have a higher crash risk than the highways and they should be used as little as possible. Therefore, the length of collector roads on a route should be minimized ([Dijkstra, 2013](#)). The percentage length share ($ls_{cr}^{p,w}$) of collector roads on a route is calculated as the length of collector roads ($l_{cr}^{p,w}$) to the total length (l_p^w) of a route p between OD pair w [Eq. \(2\)](#).

$$ls_{cr}^{p,w} = \frac{l_{cr}^{p,w}}{l_p^w} \times 100, \forall p \in P_w, w \in W \quad (2)$$

(iii). Minimize the total travel distance on a route

The total travel distance on a route explains the exposure to crash risk which is generally measured in terms of total vehicle kilometers traveled (VKT) ([Amoh-Gyimah et al., 2017](#)). Shorter the distance on a route, lesser is the exposure to crash risk. Therefore, a higher safety score requires the total distance traveled on a route to be kept to a minimum. It is calculated by adding the length of all the links passing through a route, as shown in [Equation 3](#).

$$l_p^w = \sum_{a \in A} \delta_a^p l_a, \forall p \in P_w, w \in W \quad (3)$$

(iv). Minimize the density of intersections on a route

Intersections in a road network are hotspots for serious and fatal crashes ([Alarifi et al., 2017; Huang et al., 2008](#)). In the United States, roughly around one quarter of fatal crashes and one half of serious crashes are attributed to intersections ([FHWA, 2024](#)). Thus, a route with a lower density of intersections will have a higher safety score compared to the one with a higher density. To calculate the safety score for this criterion, the intersection density is calculated as the number of intersections (n_i^p) per kilometer on a route, as shown in [Equation 4](#).

$$id_{p,w} = \frac{n_i^p}{l_p^w}, \forall p \in P_w, w \in W \quad (4)$$

2.3.2. Quantification of route safety score

For each route the safety score is quantified based on the above four criteria by using the network topology data and applying Eqs. (5) and (6). To compare the scores of different routes, min-max normalization is carried out for each criterion such that the best route is awarded a score of 0 while the worst route is awarded a score of 1. The scaled score for any other route is calculated by subtracting the lowest score from the score of the criterion in question for that specific route and dividing it by the difference between the maximum and the minimum score as shown in Equation 5 (Dijkstra, 2013; Dijkstra et al., 2007).

$$score_c^{p,w} = \frac{y_c^{p,w} - \min_c \{y_c^{p,w}\}}{\max_c \{y_c^{p,w}\} - \min_c \{y_c^{p,w}\}}, \forall p \in P_w, w \in W \quad (5)$$

.where $score_c^{p,w}$ is the normalized score for criterion c of route p and $y_c^{p,w}$ is the score of a criterion c for route p between OD pair w . The total score is calculated by multiplying the standardized score by a weight factor and added up over all criteria to get the total score indicating the level of unsafety.

$$US_p^w = \sum_{c=1}^C score_c^{p,w} \times \gamma_c, \forall p \in P_w, w \in W \quad (6)$$

.where US_p^w is the unsafety score for route p between OD pair w , γ_c is the weight factor for criterion c and C is the number of total criteria. The weight factor γ_c , in the unsafety score calculation, are set equally across the criteria (e.g., $\gamma_c = \frac{1}{|C|}$, where $|C| = 4$ in our case, yielding $\gamma_c = 0.25$ per criterion) to ensure the aggregate unsafety score remains bounded between 0 and 1 following the established literature (Dijkstra, 2013; Dijkstra et al., 2007). To arrive at the percent safety score, the unsafety score is deducted from 1 and multiplied by 100 as expressed in Equation 7.

$$\Theta_p^w = (1 - US_p^w) \times 100, \forall p \in P_w, w \in W \quad (7)$$

.where Θ_p^w is the safety score which indicates the degree of safety for each route p between OD pair w .

2.3.3. Safety-conscious route choice sets

Travelers with varying safety requirements may have different route choice sets. Some travelers might exclude certain available routes from their choice set if the safety level of those routes is less than their acceptable safety requirement. Generally, a more safety-conscious traveler will have a more limited route choice set compared to a less cautious traveler. This is because a more safety-conscious traveler may prefer to take a safer route, even if it is not the shortest option. Thus, users with heterogeneous safety requirements traveling between OD pair w can be divided into M classes, where each class has its own route choice set shown in Equation

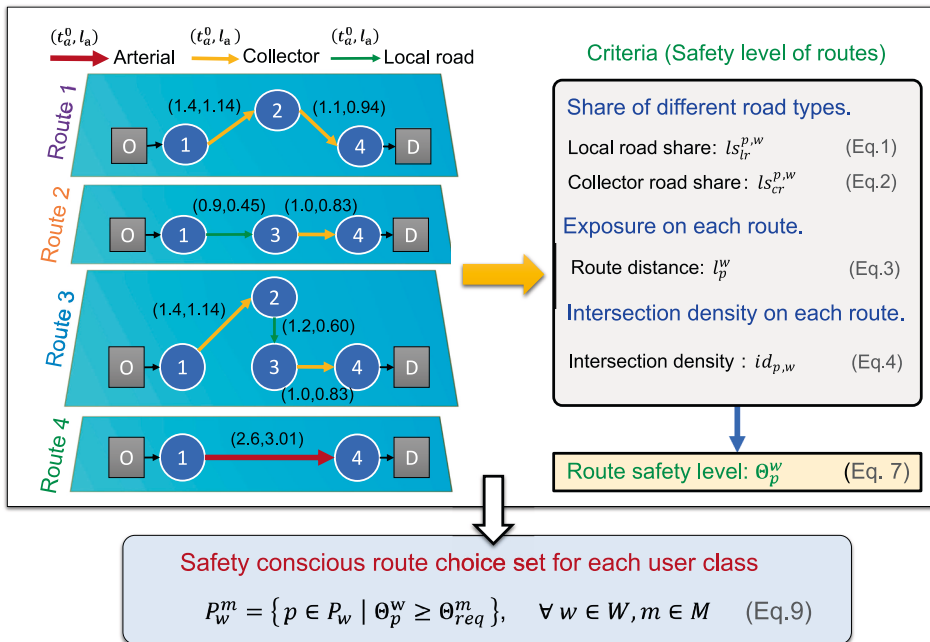


Fig. 2. Methodology for calculating safety level of routes and safety conscious route choice sets. Note. Four routes with different features are shown for illustration.

8.

$$P_w^m \subseteq P_w, \forall w \in W, m \in M \quad (8)$$

.Here P_w represents the entire route choice set while P_w^m is the route choice set for user class m between OD pair w .

Definition 1.

A route $p \in P_w$ in OD pair $w \in W$ will be included in the route choice set P_w^m for user class m if its safety score is greater than the safety requirement of that user class (Equation 9).

$$P_w^m = \{p \in P_w | \Theta_p^w \geq \Theta_{req}^m\}, \forall w \in W, m \in M \quad (9)$$

where Θ_p^w is the safety score of route p between OD pair w and Θ_{req}^m is the safety requirement of user class m . The parameter - Θ_{req}^m captures heterogeneous safety preferences across traveler classes, acting as a minimum safety threshold for route inclusion in the choice set—a concept aligned with risk-threshold approaches in prior studies (e.g., Li and Huang (2019)). While this parameter is analyst-defined in this study, it can be calibrated empirically in practice.

Fig. 2 illustrates the methodology for assessing the safety levels of routes with diverse characteristics. For demonstration, four routes with distinct features are presented. The safety level of each route is computed based on Equation 7 using the criteria in Equations (1) – (4). The safety-conscious route choice set for each user class is determined by applying Equation 9.

In Fig. 2, free-flow travel time (minutes) and travel distance (km) are displayed above each link. Each route is characterized by four criteria: local road share, collector road share, travel distance, and intersection density. Tables 3 and 4 detail the safety score calculation. Table 3 provides the unsafety scores for each criterion, while Table 4 summarizes the weighted safety scores for each route. With equal weights assigned to all criteria, the total safety score ranges from 0% to 100%.

As shown in Table 4, Route 4 attains the highest safety score ($\Theta_4 = 75\%$), whereas Route 3 has the lowest ($\Theta_3 = 32\%$). These differences arise from variations in the underlying attributes: Route 4 benefits from lower intersection density and absence of local roads, whereas Route 3 has higher shares of local roads and longer travel distance.

To illustrate how safety thresholds shape choice sets, consider two traveler classes (TC) with distinct safety requirements. TC_1 has a low safety requirement ($\Theta_{req}^{TC_1} = 0\%$), representing safety-indifferent travelers. TC_2 is more safety-conscious, with a higher threshold ($\Theta_{req}^{TC_2} = 40\%$). Applying Equation 9 based on route safety scores in Table 4 yields the following choice sets:

- Choice set for $TC_1 = \{\text{Route 1, Route 2, Route 3, Route 4}\}$.
- Choice set for $TC_2 = \{\text{Route 1, Route 4}\}$.

This example demonstrates that the choice set for a more safety-conscious class is more restricted. It also underscores that Θ_{req}^m is applied to the route-level safety score Θ_p , which is normalized and comparable across routes regardless of their link counts or topology. In practice, Θ_{req}^m should be calibrated using survey or behavioral data to reflect real-world traveler preferences.

2.4. Safety of a route based on traffic conditions

In this section, we introduce the safety of a route in relation to its traffic conditions (i.e., the travel speed and travel time). Safety of a transportation system is the product of crash risk per unit exposure and the number of exposure units occurring on the system during the specified time period as illustrated in Fig. 3 (Chapman, 1973; Hauer, 1982). The travelers crash risk is affected by a number of factors such as traffic conditions, driver behavior and environmental characteristics (e.g., weather). Exposure is defined as some form of travel either by foot or vehicle. The exposure factors measure how exposed travelers are to risk, such as traveling for longer time periods or longer distances, which could increase the likelihood of being involved in a dangerous situation. In transportation safety, travel time (Chipman et al., 1992), travel volume (Wong et al., 2007) and travel distance (Amoh-Gyimah et al., 2017) are often used as proxies for exposure. Here we define safety in terms of crash risk cost (CRC), which indicates the level of crash risk in terms of the societal losses incurred due to crash occurrence and severity. In the next section, CRC distribution is introduced to consider the reliability and unreliability of safety variability.

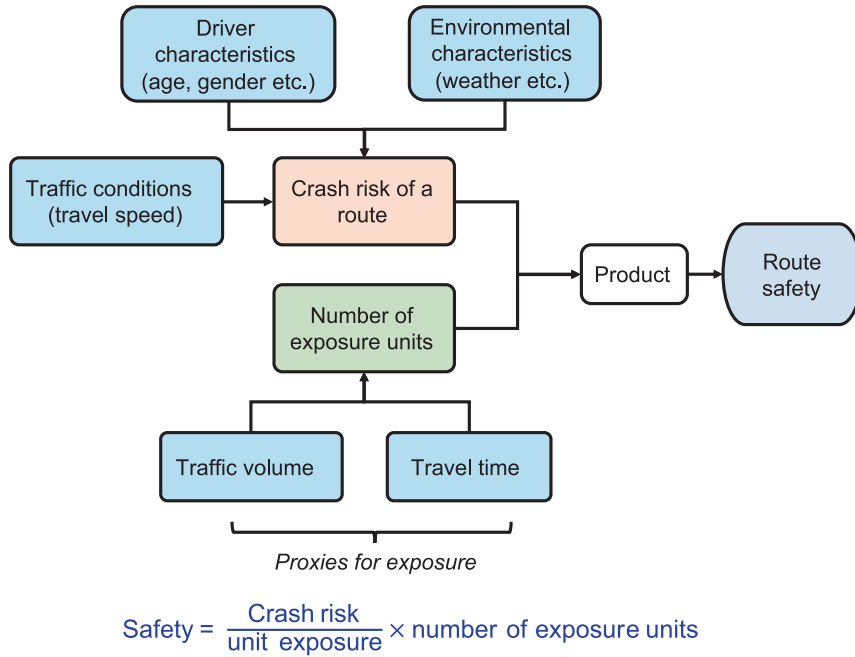
Table 3
Min-max normalized scores for each route.

Criterion	Variable	Unit	Scores for each route				–	Normalized scores for each route			
			1	2	3	4		1	2	3	4
1	ls_{lr}	%	0.00	35.1	23.3	0.00		0.00	1.00	0.67	0.00
2	ls_{cr}	%	100	64.9	76.7	0.00		1.00	0.65	0.77	0.00
3	l	km	2.08	1.28	2.57	3.01		0.46	0.00	0.74	1.00
4	id	km^{-1}	1.44	2.34	1.56	0.66		0.46	1.00	0.53	0.00

Table 4

Weighted min–max normalized scores for each route.

Criterion	Variable	Unit	Weight	Weighted normalized scores for each route			
				1	2	3	4
1	ls_{lr}	%	0.25	0.00	0.25	0.17	0.00
2	ls_{cr}	%	0.25	0.25	0.16	0.19	0.00
3	l	km	0.25	0.12	0.00	0.19	0.25
4	id	km ⁻¹	0.25	0.12	0.25	0.13	0.00
			US_p	0.48	0.66	0.68	0.25
			Θ_p	52%	34%	32%	75%

**Fig. 3.** Conceptual model of crash risk cost (safety) of a route based on traffic conditions.

2.4.1. Crash risk cost distribution

Each link in a road network has an average crash risk (average safety condition) due to its preexisting design features and traffic flow under specific conditions. The average safety condition of a road link can be estimated based on the factors related to the environment, roadway, and driving conditions (Fig. 3). However, travelers' crash risk on a road segment is random due to variability in the safety conditions. Accurately estimating the safety condition of a traveler traveling on a route may not be possible (Huang & Abdel-Aty, 2010; Mannering & Bhat, 2014). In this study, travelers are assumed to have gained knowledge of the distribution of safety variability i.e., the CRC on each road link. Travelers attain such knowledge from their long-term traveling experiences, which they incorporate into their travel decisions to reach a long-term habitual equilibrium flow pattern. Thus, in addition to the average safety condition of a road link, travelers also consider the safety reliability (probability of a trip being safe more frequently) and unreliability (probability of extremely higher CRC) aspects of safety variability.

The CRC is assumed to follow a distribution with a mean value and variance, termed CRC distribution (Huang et al., 2020). The CRC distribution for a road link (r_a) is shown in Fig. 4 (green box). μ_a is the mean of CRC distribution which corresponds to the safety aspects of road properties under specific driving conditions and σ_a^2 is the variance of CRC distribution, which represents the safety variability. Operationalizing the factors from Fig. 3, μ_a of the CRC distribution captures the systematic, flow-dependent risk determined primarily by traffic conditions. Moreover, the variance (σ_a^2) aggregates the uncertainty stemming from the stochastic influences of driver behavior, environmental conditions, and other unobserved variabilities. The hypothetical link CRC distribution illustrated in Fig. 4 serves as a theoretical model to capture the traveler's subjective perception of the mean safety and safety variability. Nonetheless, it is essential to validate the parameters of this distribution using empirical data to adopt it for different road types and traffic conditions.

To quantify the mean and variance of the CRC distribution, travel time (t_a) is used as a proxy for exposure, which can explain the variability in crash risk among drivers with different driving patterns (Pei et al., 2012). Moreover, one of the most significant factors influencing crash occurrence, crash severity, and, most importantly, traveler's safety perception is travel speed, as evident from various empirical studies (Aljanahi et al., 1999; Elvik, 2002). The relationship between the traveler's perception of mean CRC (μ_a) and

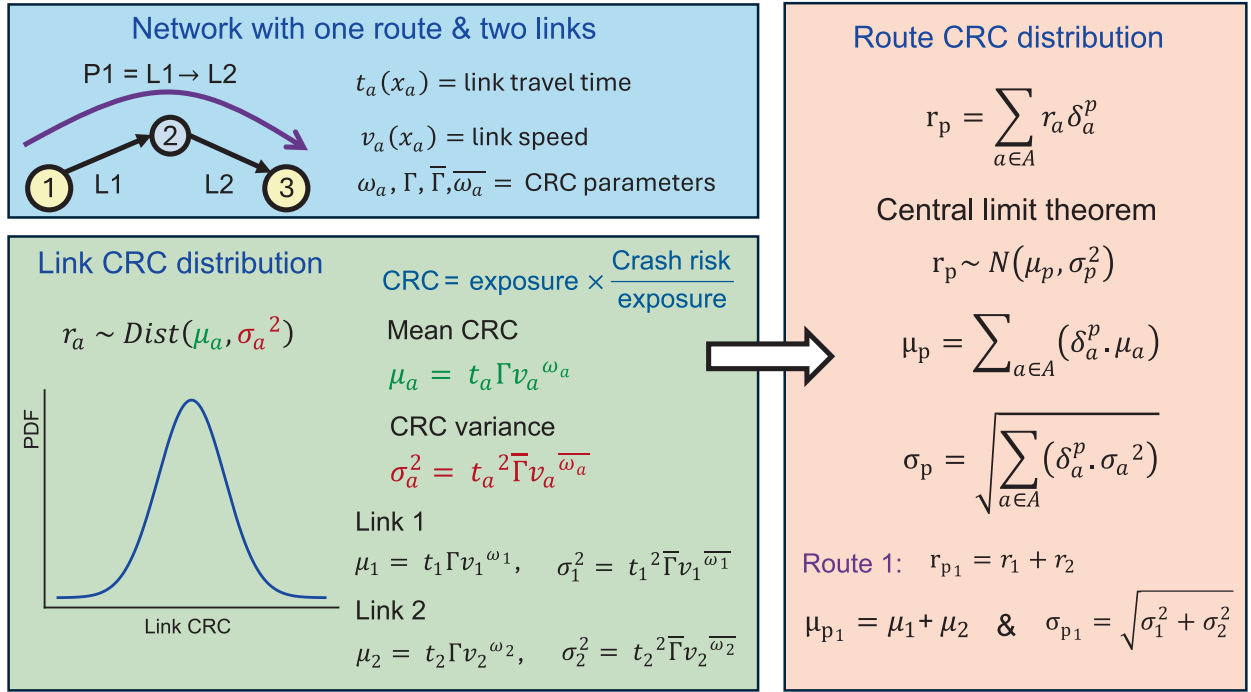


Fig. 4. Illustration of link and route CRC distribution. Note. A small network with one route and two links is shown for illustration of route CRC calculation.

the average travel speed (v_a) is based on the crash risk model proposed by Elvik et al. (2004), shown in Fig. 4 (green box). Similarly, the perceived CRC variance (σ_a^2) per unit of exposure is also assumed to be dependent on the average speed of the road segment as shown in Fig. 4 (green box) (Huang et al., 2020).

Based on the definition of link CRC distribution, the route CRC distribution (r_p) can be stated by adding the corresponding link CRC variables, as illustrated in Fig. 4 (orange box). Assuming that the link's CRC distribution is independent, bounded, and has a finite, non-zero variance, the route's CRC distribution follows a normal distribution, according to the central limit theorem (Lo et al., 2006). Therefore, the mean (μ_p) and standard deviation (σ_p) of the route's CRC can be expressed as shown in Fig. 4 (orange box). For modeling tractability and to highlight the core mechanisms of safety-oriented route choice with heterogeneous preferences, this study assumes independent link CRC distributions, allowing route CRC distributions to follow normal distribution via the central limit theorem. This mirrors common assumptions in travel time variability models (Chen & Zhou, 2010; Lam et al., 2008; Lo et al., 2006; Shao et al., 2006), despite empirical evidence of link correlations due to congestion propagation, shared infrastructure, or external factors such as weather (Liu et al., 2017; Prakash & Srinivasan, 2015; Srinivasan et al., 2014). While such correlations have been addressed in advanced equilibrium frameworks (Chen et al., 2020; Prakash et al., 2018), their incorporation here is left for future work to maintain focus on behavioral insights.

Additionally, a small network example with two links and one route, depicted in Fig. 4 (blue box), demonstrates how the route CRC distribution is derived from the link CRC distributions, as shown in Fig. 4 (orange box). In the next section, we explain the concept of mean excess crash risk cost (MECRC).

2.4.2. Definition of mean excess crash risk cost

Various aspects of the transportation system are unreliable: for example, whether an individual will be able to arrive at the destination on time, whether an individual will be able to get a seat on the bus, and whether an individual will be able to reach his destination safely (Bates et al., 2001). In theory, how individual choices are made relies on the generalized cost, which is directly related to utility theory. Under uncertainty: an individual chooses an option, keeping in mind the probability of different outcomes, that has the highest value of expected utility (minimum cost) and minimum associated risk/uncertainty (developed from the basic theory proposed by Von Neumann and Morgenstern (1944)) (Bates et al., 2001). The expected value of choice is represented by the mean and the risk or uncertainty is represented by the variance.

This study assumes that travelers have gained prior knowledge of the distribution of each alternative's safety variability (CRC variability) based on their past traveling experiences. Thus, travelers incorporate this information into their route choice decisions along with their safety preferences to reach a long-term habitual flow pattern. Travelers have different safety preferences as some may be more concerned about their safety than others, judged by their driving behaviors (May et al., 2008). For example, it has been found that old drivers are more safety-conscious than young drivers (Castellà & Pérez, 2004; Deery, 1999; McKenna & Horswill, 2006). By

considering the safety reliability requirements, travelers search for a route such that the corresponding effective CRC ensures a higher probability of safe arrival with a predefined confidence level α (Huang et al., 2020), as shown in Equations 10 and 11.

$$R_p^w(\alpha) = \min \left(R \mid \Pr(S_p^w \leq R) \geq \alpha \right), \quad \forall p \in P_w, w \in W \quad (10)$$

$$R_p^w(\alpha) = \mu_p^w + \phi(\alpha) \cdot \sigma_p^w, \quad \forall p \in P_w, w \in W \quad (11)$$

where R is the effective CRC required to ensure safe arrival at a confidence level α and $\phi(\alpha) \cdot \sigma_p$ is the extra CRC added to the mean CRC as CRC margin/buffer CRC, to ensure more frequent safe arrivals under the CRC reliability requirement at a predefined confidence level α . This concept is analogous to the concept of travel time budget (Lo et al., 2006).

However, travelers with higher safety concerns also consider the consequences of excessively higher crash risk costs (i.e., the unreliability aspects of safety variability). If the consequences of traveling on a particular route result in a very high CRC, such as a traffic crash with a higher probability of severe injury or fatality, the route would be considered entirely unacceptable to the traveler. The CRC is associated with various factors, including driving conditions, roadway features, and human behavior. Driving conditions, such as heavy traffic and high travel speeds, can increase the risk of crashes, as can poorly designed roadway features such as undivided highways or roads with poorly designed horizontal and vertical curves. Studies have shown that impact speed has a direct relationship with crash severity (Hauer, 2009; Solomon, 1974), and heavy-traffic volume increases the probability of severe crashes (Behnood & Mannering, 2019). Additionally, traffic crashes on undivided highways have a higher likelihood of resulting in severe injury or fatality (Kadeha et al., 2021; Kidando et al., 2019). Thus, it is reasonable for travelers to choose a route that ensures safety reliability (defined by effective CRC) more frequently while minimizing the expected unreliability aspect (i.e., extremely high CRC beyond effective CRC). It means travelers ensure a safe trip more frequently at a specified confidence level α , and avoid the routes with a higher probability of fatal/severe crashes (extremely high CRC). This tradeoff between the reliable and unreliable aspects of the safety variability can be modeled by adopting the METE model (Chen & Zhou, 2010; Chen et al., 2011; Xu et al., 2023; Xu et al., 2014). The concept of mean excess crash risk cost (MECRC) is explained here.

The mean excess CRC, $\psi_p^w(\alpha)$, for a route $p \in P^w$ between OD pair w at a specified confidence level α is equal to the conditional expectation of CRC exceeding the corresponding route's effective CRC, $R_p^w(\alpha)$, as shown in Equation 12.

$$\psi_p^w(\alpha) = E \left[S_p^w \mid S_p^w \geq R_p^w(\alpha) \right], \quad \forall p \in P_w, w \in W \quad (12)$$

where S_p^w is the random CRC on a route p between OD pair w , $E[\cdot]$ is the expectation operator, and R_p^w is the effective CRC defined in Equation 11.

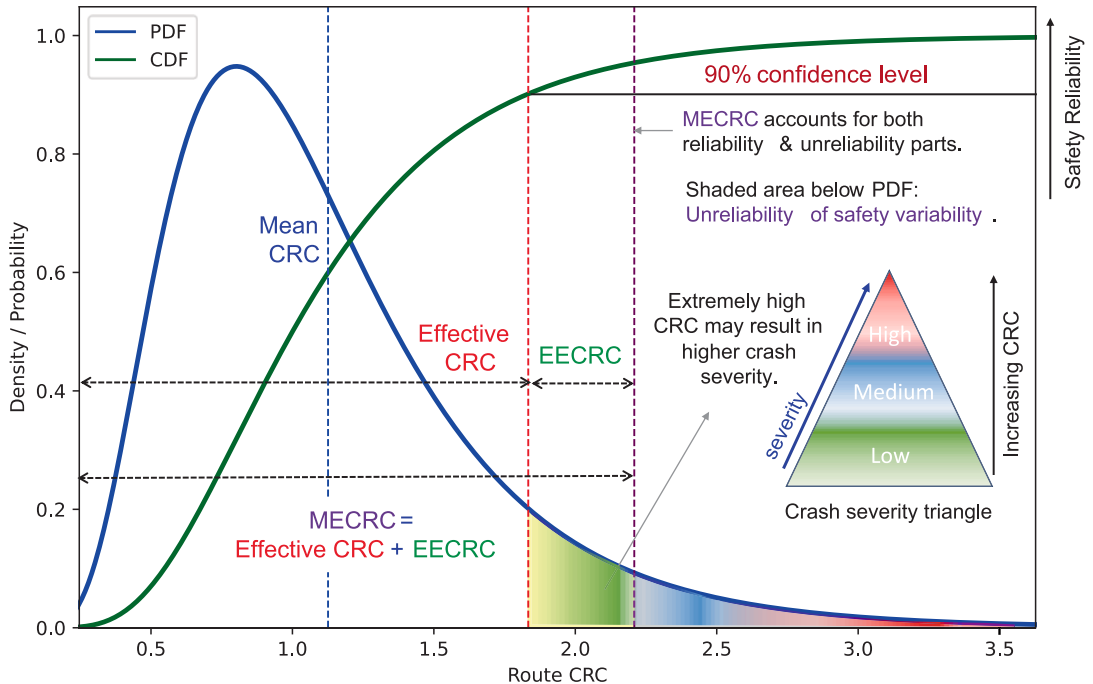


Fig. 5. Illustration of the mean excess crash risk cost (MECRC).

If the route CRC distribution $f(S_p^w)$ is known, the mean excess CRC (MECRC) definition in Equation 12 as a conditional expectation can be rewritten as:

$$\psi_p^w(\alpha) = E[S_p^w | S_p^w \geq R_p^w(\alpha)] = \int_{S_p^w \geq R_p^w(\alpha)} \frac{S_p^w f(S_p^w)}{\Pr(S_p^w \geq R_p^w(\alpha))} d(S_p^w) \quad (13)$$

$$\psi_p^w(\alpha) = \frac{1}{1-\alpha} \int_{S_p^w \geq R_p^w(\alpha)} S_p^w f(S_p^w) d(S_p^w) \quad (14)$$

Equation 14 can be written as.

$$\psi_p^w(\alpha) = R_p^w(\alpha) + E(S_p^w - R_p^w(\alpha) | S_p^w \geq R_p^w(\alpha)) \quad (15)$$

The MECRC can be decomposed into two parts. The first part is exactly the effective CRC of route p , explaining the reliability aspect of the safety variability (more frequent safe arrivals) at a user-specified confidence level α . The second part is the “expected excess CRC (EECRC)” beyond the effective CRC, reflecting the unreliability aspect of safety variability (i.e., possible high-risk scenarios) in the distribution tail of $1-\alpha$. MECRC criterion based on the METE model, incorporates the safety variability's reliability and unreliability aspect. Both aspects are assumed to affect the traveler's route choice decisions in our model.

To explain the concept of MECRC, Fig. 5 shows an illustration of a route's hypothetical CRC distribution, with both the probability density function (PDF represented by a blue line) and cumulative density function (CDF represented by a green line). At a specified confidence level α , effective CRC is the minimum CRC threshold, which is the sum of mean CRC (MCRC) and buffer CRC (BCRC), accepted by the travelers such that the corresponding cumulative probability of actual CRC less than this limit is at least α . The tail (shaded area) of the distribution beyond effective CRC represents all the possible high-risk scenarios (potential crashes) in which the actual CRC is higher than the effective CRC. In the case of unreliability aspect of safety variability, travelers allow a higher CRC margin to further increase their probability of safe arrival. It is evident from the Fig. 5 that effective CRC does not assess the magnitude of crash risk cost associated with the worst situations and is unable to distinguish between the situations where the actual CRC is only a little higher (low severity crash) than effective CRC from those in which the actual CRC is extremely higher (high severity crash). Therefore, the effective CRC accounts for the reliability aspect of safety variability at a specified confidence level α , while MECRC accounts for both the reliability of safety variability (i.e., effective CRC required for safe arrival at a confidence level α) and the unreliability of safety variability (i.e., accounting for the worst crash severity levels). The MECRC consists of both effective CRC and the expected excess CRC beyond the effective CRC.

This study assumes that the link crash risk costs are independent. A similar assumption was adopted by Huang et al. (2020) to model the safety reliability. Thus, according to the central limit theorem, the route CRC distribution follows a normal distribution, as shown in Fig. 4. According to the definition of MECRC in Equation 12, the route MECRC can be expressed as the following minimization program.

$$\min \frac{1}{1-\alpha} \int_{R_p^w}^{\infty} S_p^w \cdot \frac{1}{\sigma_p^w \sqrt{2\pi}} \exp\left(-\frac{(S_p^w - \mu_p^w)^2}{2(\sigma_p^w)^2}\right) d(S_p^w) \quad (16)$$

$$\text{s.t.} \int_{-\infty}^{R_p^w} \frac{1}{\sigma_p^w \sqrt{2\pi}} \exp\left(-\frac{(S_p^w - \mu_p^w)^2}{2(\sigma_p^w)^2}\right) d(S_p^w) \geq \alpha, \quad (17)$$

where Equation 17 represents the crash risk cost reliability chance constraint. MECRC of route p can be represented as.

$$\psi_p^w = \mu_p^w + \frac{\sigma_p^w}{\sqrt{2\pi(1-\alpha)}} \exp\left(-\frac{(\phi^{-1}(\alpha))^2}{2}\right) \quad (18)$$

We assume that the link crash risk costs are independent, however in reality they may not be independent due to the network topology or sources of uncertainty and thus link interactions should be considered in future research.

2.5. User equilibrium formulation considering safety-conscious route choice sets

In this section, we provide the user equilibrium formulation for the METE model considering both the travelers' travel time variability and safety variability concerns. In general, traveler classes characterized by different levels of safety variability concerns and a corresponding confidence level (α_m) exist on the road network. Each class m has its own route choice set ($P_w^m \subseteq P_w$) determined by its specific safety requirements, denoted by Θ_{req}^m as shown in Equation 9. Thus, the generalized cost for class m can be represented as:

$$G_{p,m}^w = \psi_{p,m}^w + \Lambda(\eta_p^w) \quad (19)$$

where $\psi_{p,m}^w$ is the MECRC for travelers of class m (Equation 20).

$$\psi_{p,m}^w = \mu_p^w + \frac{\sigma_p^w}{\sqrt{2\pi}(1 - \alpha_m)} \exp\left(-\frac{(\phi^{-1}(\alpha_m))^2}{2}\right) \quad (20)$$

η_p^w is the travelers' METT and Λ is the cost-converting factor of travel time. METT is also assumed to follow normal distribution and can be expressed as follows (Chen & Zhou, 2010):

$$\eta_p^w = T_p^w + \frac{\widetilde{\sigma}_p^w}{\sqrt{2\pi}(1 - \widetilde{\alpha})} \exp\left(-\frac{(\phi^{-1}(\widetilde{\alpha}))^2}{2}\right) \quad (21)$$

It is reasonable to assume that travelers of each class are willing to minimize their generalized cost when traveling between an OD pair w in an uncertain environment. Accordingly, the long-term habitual flow pattern can be reached, which is termed as multiclass mean excess traffic equilibrium (MC-METE) model. Let G represent the generalized cost vector $(\dots, G_{p,m}^w, \dots)^T$, π_m^w represent the minimum generalized cost for traveler class m between an O-D pair w , i.e., $\pi_m^w = \min\left(\sum_{m=1}^M \sum_{p \in P_w} G_{p,m}^w\right)$, and f denote the route-flow vector $(\dots, f_{p,m}^w, \dots)^T$. Thus, the equilibrium conditions can be stated as follows.

At the equilibrium state, for each traveler class, no traveler can improve his/her generalized cost by unilaterally changing routes. It means that all the routes used between an OD pair w have equal and minimal generalized cost and no unused route has a lower generalized cost than the routes considered in the choice set of each user class m , i.e., the following conditions hold (Equation 22).

$$G_{p,m}^w(f^*) - \pi_m^w \begin{cases} = 0 & \text{if } (f_{p,m}^w)^* > 0 \\ \geq 0 & \text{if } (f_{p,m}^w)^* = 0 \end{cases}, \forall p \in P_w^m, \forall w \in W, \forall m \in M \quad (22)$$

Such an equilibrium state can be formulated as a variational inequality (VI) problem, where we want to find a solution vector of route flows $f^* \in \Omega$, such that:

$$G(f^*)^T(f - f^*) \geq 0, \forall f \in \Omega \quad (23)$$

where, the feasible set Ω is defined by the following Equations:

$$\sum_{p \in P_w^m} f_{p,m}^w = q_w^m, \forall w \in W, \forall m \in M \quad (24)$$

$$x_a = \sum_{m \in M} \sum_{w \in W} \sum_{p \in P_w^m} f_{p,m}^w \delta_a^p, \quad \forall a \in A \quad (25)$$

$$f_{p,m}^w \geq 0, \forall p \in P_w^m, \forall w \in W, \forall m \in M \quad (26)$$

where $f_{p,m}^w$ is the flow for traveler class m on route $p \in P_w^m$ connecting an OD pair w ; $G_{p,m}^w$ is the generalized travel cost for traveler class m on route p connecting an OD pair w ; q_w^m is the travel demand for traveler class m between an OD pair w ; x_a is the link flow on the link a ; δ_a^p is the link-route incidence indicator: $\delta_a^p = 1$, when link a is on route p connecting an OD pair w , and zero otherwise. The conservation, definitional, and non-negative constraints are given in Equations (24), (25), and (26), respectively.

The following two propositions give the equivalence of the VI formulation and the equilibrium state as well as the existence of an equilibrium solution.

Proposition 1.

(Equivalence). Assume that the route generalized cost function $G(f)$, which is a summation of the route's mean excess travel time function and mean excess crash risk cost function, is positive, the solution of the VI problem is equivalent to the equilibrium solution of the MC-METE model.

Proof. Note that f^* is a solution of the VI problem if and only if it is a solution of the following linear program:

$$\min_{f \in \Omega} G(f^*)^T f \quad (27)$$

By considering the primal–dual optimality conditions of (27), we have.

$$f_{p,m}^{w*} (G_{p,m}^w(f^*) - \pi_m^w) = 0, \forall p \in P_w^m, \forall w \in W, \forall m \in M \quad (28)$$

$$G_{p,m}^w(f^*) - \pi_m^w \geq 0, \forall p \in P_w^m, \forall w \in W, \forall m \in M \quad (29)$$

It is easy to see that the MC-METE condition (22) is satisfied. This completes the proof. $\square$

Proposition 2.

(Existence). The proposed VI problem has at least one solution under the assumption that the generalized travel cost on each route between O-D pair w for each traveler class is positive and continuous with respect to its route flow.

Proof. Note that the feasible set Ω is nonempty, closed, and convex. Moreover, the mapping $G(f)$ is positive and continuous with respect to f . Thus, the VI problem admits at least one solution. This completes the proof. $\square$

3. Solution algorithm

This section introduces the two-stage algorithm for solving the traffic assignment model. In Stage 1, safety scores for all routes in the given route set between each OD pair are calculated using the methodology in Section 2.3. Based on these scores, route choice sets are determined for each class according to their safety score requirements. In Stage 2, the non-additive traffic assignment problem is

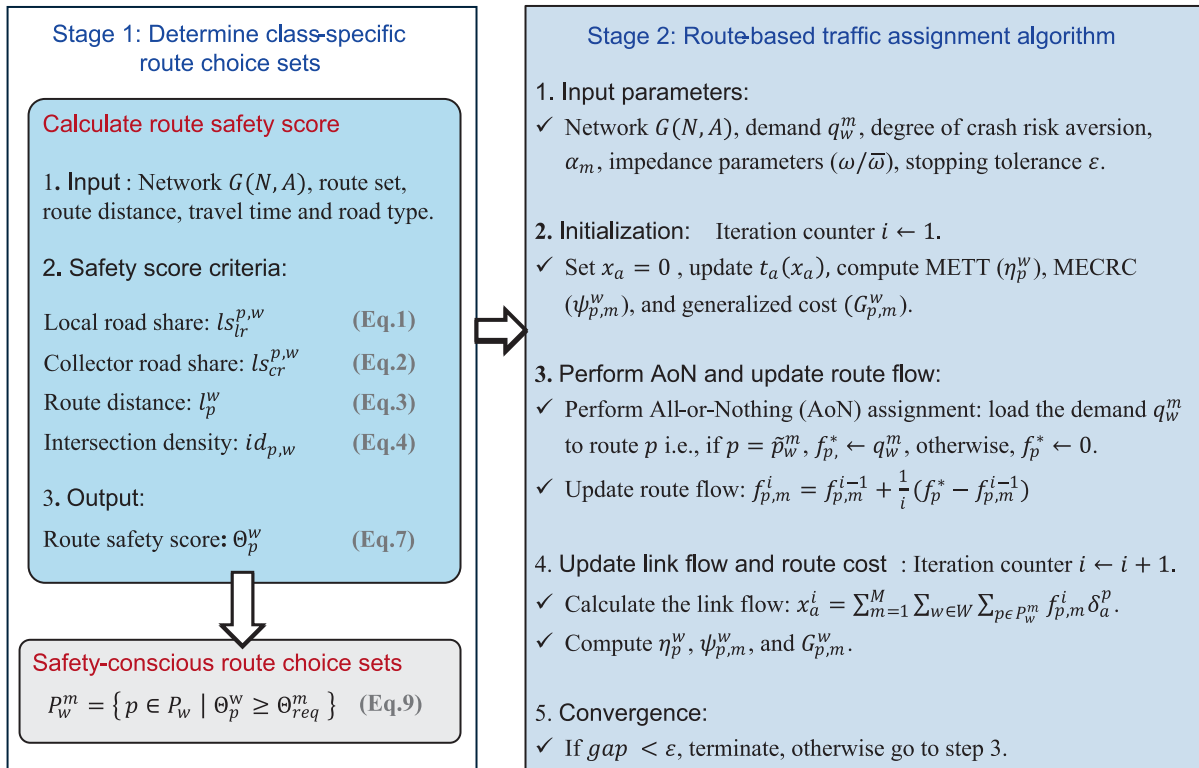


Fig. 6. An overview of the two-stage algorithm.

solved using a route-based algorithm, as the route costs are non-additive with link costs and must be treated as route-specific (Lo & Chen, 2000). The method of successive averages (MSA) is used to iteratively update the step size and flow. An overview of the algorithm is shown in Fig. 6.

4. Numerical experiments

This section presents the results from two networks to demonstrate the properties and application of the model. First, we analyze a small network with two traveler classes to illustrate the safety-conscious route choice behavior. This is followed by an examination of a real-world network, specifically the Winnipeg network.

4.1. A small network

In this example, a small network consisting of 6 links and 4 routes is illustrated in Fig. 7. The link free-flow travel time, link capacity, link length, variance of free-flow travel time, and travelers' risk perception of a link related to its features are shown in Table 5. The cost converting factor " Λ " is set equal to 3.0. The CRC and travel time for each link are assumed to follow a normal distribution for the ease of using the analytical expressions of the MECRC and METT for each route. The mean travel time is calculated using the well-known BPR function provided in Eq. (30).

$$t_a = t_a^0 \left(1 + 0.15 \left(\frac{x_a}{c_a} \right)^4 \right) \quad (30)$$

4.1.1. Single class travelers to compare different traffic assignment models

For the comparison among different traffic assignment models, we consider a single user class with the safety score requirement $\Theta_{req} = 0\%$, meaning all the routes are included in the choice set. The confidence levels for both the CRC distribution (α) and the travel time distribution ($\tilde{\alpha}$) are set equal to 0.9. The total demand is set at 3000 pcu/h. The results are presented in Fig. 8, which illustrates how equilibrium route flows change under different route choice criteria.

In the UE case shown in Fig. 8(a), the highest flow is noticed on Route 2, followed by Route 4. Fig. 8(b) presents the METE model in which travelers consider mean excess travel time (METT) only; here, flows from Routes 1 and 3 shift to Route 4 due to its lowest travel time variability, reflected in the lowest variance ($\tilde{\sigma}_4 = 1.0$). Fig. 8(c) shows the METE model where travelers consider both METT and mean excess crash risk cost (MECRC). In this case, Route 4 attracts no flow because of its high CRC variability ($\sigma_p = 4.38$), and high mean crash risk ($\mu_4 = 3.80$), which together yield a very high MECRC ($\psi_4 = 11.49$). Although Route 4 has a relatively low METT ($\eta_4 = 4.34$) due to shorter mean travel time ($T_4 = 2.58$) and low travel time variability ($\tilde{\sigma}_4 = 1.0$), the high MECRC substantially increases its generalized cost compared to Routes 1 and 2 at equilibrium. Consequently, Route 4 remains unused in Fig. 8(c), whereas in Fig. 8(a) and Fig. 8(b) it attracts demand because its cost is same as Routes 1 and 2 under those criteria.

This comparison demonstrates how equilibrium flow patterns differ depending on whether travelers account for expected travel time alone, travel time variability, or both travel time and safety variability in their generalized cost.

4.1.2. Multiclass travelers with heterogeneous safety concerns

To model travelers with heterogeneous safety concerns, we assume that the network consists of two traveler classes (TC) in a multiclass mixed equilibrium model. Each TC accounts for 50% of the total demand and the total demand is 3000 pcu/h. The confidence level $\tilde{\alpha}$ for travel time distribution equals 0.9 for both the classes. TC₁ is low safety-conscious while TC₂ is high safety-conscious. We look into safety heterogeneity from two perspectives: (a) safety-conscious class-specific route sets by setting different Θ_{req}^m values and (b) heterogeneous safety variability preferences by setting different α_m values for the CRC distribution. Each class is assumed to have its own route choice set based on the safety score requirement (Θ_{req}^m) of each class (Table 6). The safety score requirement of TC₁ ($\Theta_{req}^{TC_1}$) is set to 0% in all cases, meaning that all routes are included in its choice set. On the other hand, TC₂, which

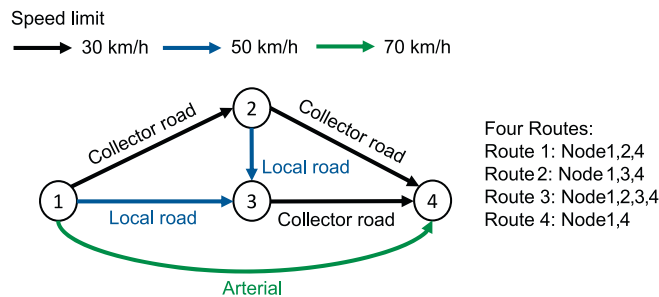
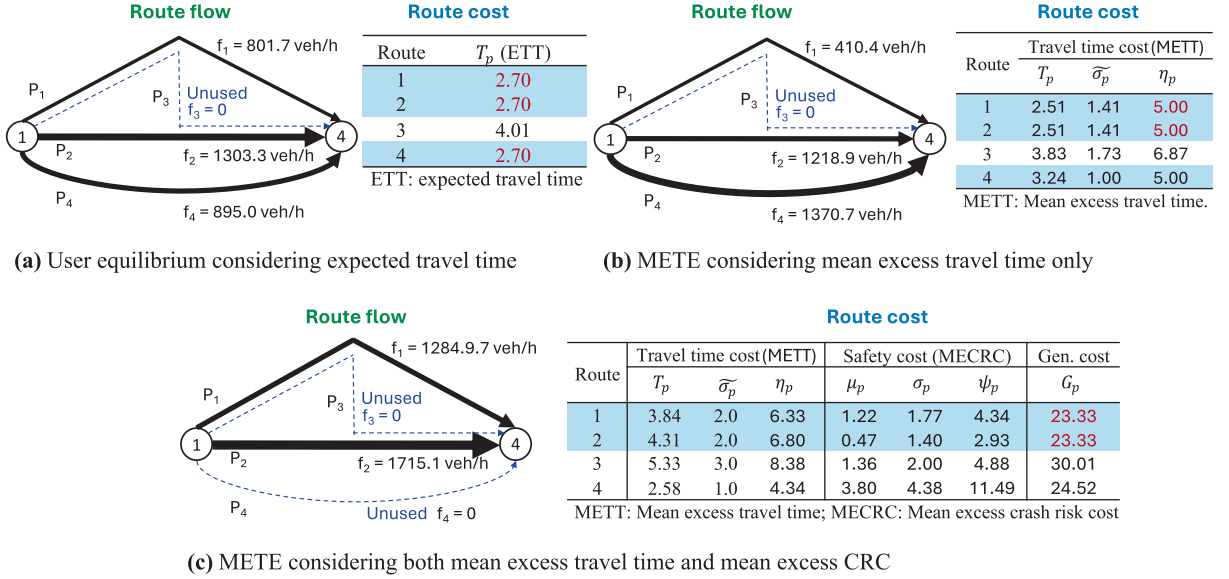


Fig. 7. A small network comprising four routes, each characterized by distinct road functional classes.

Table 5

Link characteristics for a small network.

Link	Type	t_a^0 (min)	C_a (pcu/h)	l (km)	$\tilde{\sigma}_a^2$ (min)	ω_a	$\bar{\omega}_a$
1–2	Collector	1.4	921	1.14	1.0	2.0	2.5
1–3	Local road	0.9	952	0.45	1.0	2.1	2.6
1–4	Arterial	2.6	1200	3.01	1.0	2.0	2.5
2–3	Local road	1.2	921	0.60	1.0	2.0	2.5
2–4	Collector	1.1	952	0.94	1.0	2.0	2.5
3–4	Collector	1.0	1071	0.83	1.0	2.1	2.6

**Fig. 8.** Equilibrium path flow patterns for different traffic assignment models.**Table 6**

Cases considered in numerical experiments.

Case	Safety score requirement for each class		Generalized travel cost
	$\Theta_{req}^{TC_1}$	$\Theta_{req}^{TC_2}$	
1	0%	0%	METT and MECRC
2	0%	40%	METT and MECRC
3	0%	60%	METT and MECRC
4	0%	0%	ETT and MECRC

Note: MECRC stands for “mean excess crash risk cost”; ETT stands for “expected travel time”; and METT stands for “mean excess travel time”.

is considered more safety-conscious, has a different safety score requirement ($\Theta_{req}^{TC_2}$) for each case (Table 6). The confidence level α_m for the CRC distribution equals 0.5 for TC_1 while it is equal to 0.9 for TC_2 , which means TC_2 is more concerned about CRC variability.

Fig. 9 presents a comparison of the unsafety scores for each criterion and total safety score across three routes. Route 4 has the highest safety score ($\Theta_4 = 75\%$), as it has the lowest unsafety scores for three (i.e., ls_r , ls_{cr} and id) out of four criteria. Routes 1 and 4 have safety scores greater than 50% while routes 2 and 3 have safety scores less than 50%.

4.1.2.1. Equilibrium results for travelers with heterogeneous safety concerns (Case 1). The results for the multiclass mixed equilibrium model for case 1 are presented in Table 7. In this case, all the routes are considered in the choice sets of both classes. Route 2 stands out as the safest option based on MECRC, exhibiting both the smallest mean CRC ($\mu_2 = 0.75$) and the lowest CRC variability ($\sigma_2 = 1.25$). It also has the lowest MECRC ($\psi_2 = 1.73$), while route 4 achieves the lowest METT ($\eta_4 = 4.34$) for both the traveler classes. For TC_1 , which is low risk-averse, most of the flow is allocated to route 1, which has the second-lowest MECRC while a portion of flow is also assigned to route 2, which has the lowest MECRC, reflecting some preferences for safer routes. Additionally, some flow is also allocated to route 4 due to its lowest METT among all the routes, attributed to its minimal travel time variability. At equilibrium, it is observed that all the three used routes (routes 1, 2, and 4) have the same generalized cost ensuring consistency with equilibrium conditions.

For the TC_2 , which is high safety-conscious (higher concern for safety variability), all the flow is allocated to route 2, as it has the

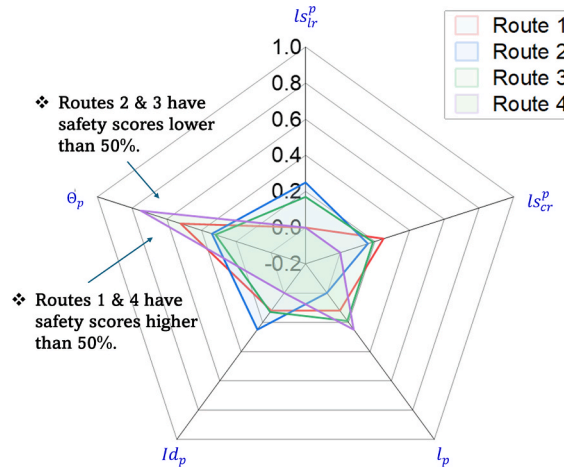


Fig. 9. Illustration of unsafety scores for each criterion and total safety score for routes (safety scores are expressed in %, e.g., 0.5 = 50%).

Table 7

Equilibrium flow of the multiclass travelers for case 1 ($\Theta_{req}^{TC_1} = \Theta_{req}^{TC_2} = 0\%$).

Class	Route	η_p	μ_p	σ_p	ψ_p	G_p	f_p	Used route
TC ₁	1	5.81	1.41	1.84	2.89	20.32	1136.96	✓
	2	6.20	0.75	1.25	1.73	20.32	96.75	✓
	3	7.83	1.69	1.95	3.24	26.73	0.00	×
	4	4.34	3.80	4.38	7.29	20.32	266.28	✓
TC ₂	1	5.81	1.41	1.84	4.65	22.07	0.00	×
	2	6.20	0.75	1.25	2.94	21.53	1500.00	✓
	3	7.83	1.69	1.95	5.12	28.61	0.00	×
	4	4.34	3.80	4.38	11.49	24.52	0.00	×

Note: ✓ denotes used route, × denotes unused route.

lowest MECRC and the smallest CRC variability. No flow is observed on the other three routes. This result highlights that the high safety-conscious travelers in TC₂ exclusively choose route 2, prioritizing its lowest MECRC and CRC variability while avoiding the other three routes with a higher CRC variability. These travelers aim to ensure a safe arrival and avoid routes with a higher likelihood of severe or fatal crashes.

4.1.2.2. Effect of different route choice sets (Cases 2–3). This section presents the effect of different route choice sets on the equilibrium flow pattern in multiclass mixed equilibrium model for cases 2 and 3. For case 2, $\Theta_{req}^{TC_2}$ is equal to 40%, therefore, the routes with the safety score of less than 40% (shown in Fig. 9) are not considered as part of the choice set by TC₂. Only routes 1 and 4 are considered in the choice set by TC₂ while TC₁ considers all the 4 routes. For TC₂ most of the flow is allocated to route 1 which has the lowest MECRC between the two considered routes while the flow of TC₁ is all allocated to route 2 with the lowest MECRC (Table 8). We also notice that the generalized cost for route 2 is lowest for TC₂, however no flow is allocated to this route as it is not considered as part of the choice set. This finding shows that a more safety conscious traveler class may be willing to travel on routes with a higher generalized cost if their safety scores are lower than the safety requirement of that class.

Table 9 presents the case in which the safety requirement is equal to 60% for TC₂ thus only route 4 is considered as part of the

Table 8

Equilibrium flow of the multiclass travelers for case 2 ($\Theta_{req}^{TC_2} = 40\%$).

Class	Route	η_p	μ_p	σ_p	ψ_p	G_p	f_p	Used route
TC ₁	1	6.78	1.09	1.72	2.47	22.81	0.00	×
	2	5.80	0.84	1.29	1.87	19.26	1500	✓
	3	8.23	1.58	1.91	3.10	27.79	0.00	×
	4	4.34	3.80	4.38	7.29	20.32	0.00	×
TC ₂	1	6.78	1.09	1.72	4.16	24.50	1381.67	✓
	2	5.80	0.84	1.29	3.10	20.49	0.00	NC
	3	8.23	1.58	1.91	4.93	29.62	0.00	NC
	4	4.34	3.80	4.38	11.47	24.50	118.33	✓

Note: ✓ denotes used route, × denotes unused route, NC denotes “not considered” route.

Table 9Equilibrium flow of the multiclass travelers for case 3 ($\Theta_{req}^{TC_2} = 60\%$).

Class	Route	η_p	μ_p	σ_p	ψ_p	G_p	f_p	Used route
TC ₁	1	4.99	1.87	1.98	3.45	18.41	93.68	✓
	2	5.48	0.93	1.32	1.98	18.41	1406.32	✓
	3	7.06	2.09	2.07	3.74	24.92	0.00	×
	4	5.29	2.78	4.05	6.01	21.88	0.00	×
TC ₂	1	4.99	1.87	1.98	5.35	20.31	0.00	NC
	2	5.48	0.93	1.32	3.26	19.68	0.00	NC
	3	7.06	2.09	2.07	5.72	26.90	0.00	NC
	4	5.29	2.78	4.05	9.90	25.76	1500.00	✓

Note: ✓ denotes used route, × denotes unused route, NC denotes “not considered” route.

choice set and all the flow is allocated to this route while ignoring the other 3 routes. This route has a higher generalized cost compared to routes 1 and 2, however those routes are not considered in the choice set due to their safety score being lower than the safety requirement of TC₂. Fig. 10 presents the summary and comparison of the results for the first three cases presented in Tables (7) - (9)..

Fig. 11 illustrates the number of routes considered in the choice of TC₂ under different safety score requirements ($\Theta_{req}^{TC_2}$). At a higher value of $\Theta_{req}^{TC_2}$, there are a smaller number of routes in the choice set, resulting in an overall increase in the network generalized travel cost.

4.1.2.3. Expected travel time versus METT in generalized cost (Case 4). This section presents and compares the results for the case where the expected travel time is considered in the generalized cost instead of the METT. Differences in the flow patterns are observed between the two cases for TC₁. When the expected travel time is considered (Table 10), the flow is allocated exclusively to routes 1 and 2 for TC₁, as they have lower generalized costs. Both routes also exhibit the lowest MECRC compared to the other routes, even though their expected travel times are higher than that of route 4. In contrast, when travel time variability is considered in case 1 (Table 7), a portion of the flow is allocated to route 4. This is due to its lower travel time variability ($\sigma_4 = 1.0$), which results in the lowest METT among all routes, despite having the highest MECRC. Due to the consideration of travel time variability, travelers of class 1 also allocate some flow to route 4, as its generalized cost becomes equal to that of routes 1 and 2. This is because routes 1 and 2 have higher travel time variability, while route 4 offers lower variability, making it an attractive option despite its higher MECRC.

This finding demonstrates that equilibrium flow patterns vary depending on whether travelers consider expected travel time or METT along with MECRC when making route choice decisions. Incorporating both METT and MECRC in the generalized cost highlights the trade-offs between minimizing the travel time variability and CRC variability in travelers route choice.

4.2. Real world network (Winnipeg network)

In this section, the proposed model and solution procedure are demonstrated using the Winnipeg network (Fig. 12), which is a real-scale network with 2535 links and 4345 OD pairs. The total demand on the network is 54,459 trips. The working route set of the network is from Bekhor et al. (2008), where the routes were generated by using a combination of the link elimination method and penalty method. The total number of routes for all OD pairs is 174,491 (an average of 40.1 routes per OD pair). The maximum number of routes generated for any OD pair is 50.

4.2.1. Experimental setting for the Winnipeg network

Since the original network does not include road functional class information, we utilized Google Maps to classify roads into two types: arterials and non-arterials (collectors/local roads). Consequently, only three criteria are used to determine the safety score of each route. The variance-to-mean ratio of travel time is assumed to be equal to 0.3. The values of ω and $\bar{\omega}$ for all the links are set equivalent to 2.1 and 2.5, respectively. The adjustment coefficients used in the unit crash risk and safety reliability functions explained in Fig. 4 are taken as $\Gamma = 3 \times 10^{-4}$ and $\bar{\Gamma} = 7 \times 10^{-5}$ respectively.

The confidence level α_m for the CRC distribution equals 0.5 for TC₁ while it is equal to 0.95 for TC₂. The confidence level $\tilde{\alpha}$ for travel time distribution equals 0.9 for both the classes. Moreover, each class is assumed to have its own route choice set based on the safety score requirement (Θ_{req}^m). $\Theta_{req}^{TC_1}$ and $\Theta_{req}^{TC_2}$ are set equal to 0% and 50% respectively, thus TC₂ only considers the routes with a safety score of greater than 50% in its choice set.

The route-based traffic assignment algorithm is tested on a PC with intel(R) Core (TM) i7-10700 CPU 2.9 GHz and 16 GB RAM using the Windows 10 operating system. The stopping criteria are set equal to 1e-4.

4.2.2. Equilibrium results for multiclass travelers with heterogeneous safety concerns

The aggregate link flows for the equilibrium model are presented in Fig. 13, which highlights the distribution of flows across the network, with notably higher volumes observed in the CBD area. To further investigate the route choice behavior of heterogeneous travelers, we conducted a disaggregate analysis focusing on two high-demand OD pairs: (92, 4) and (79, 103). We consider two traveler classes, where TC₁ exhibits a lower crash risk aversion whereas TC₂ shows higher aversion to crash risk. Only the routes that

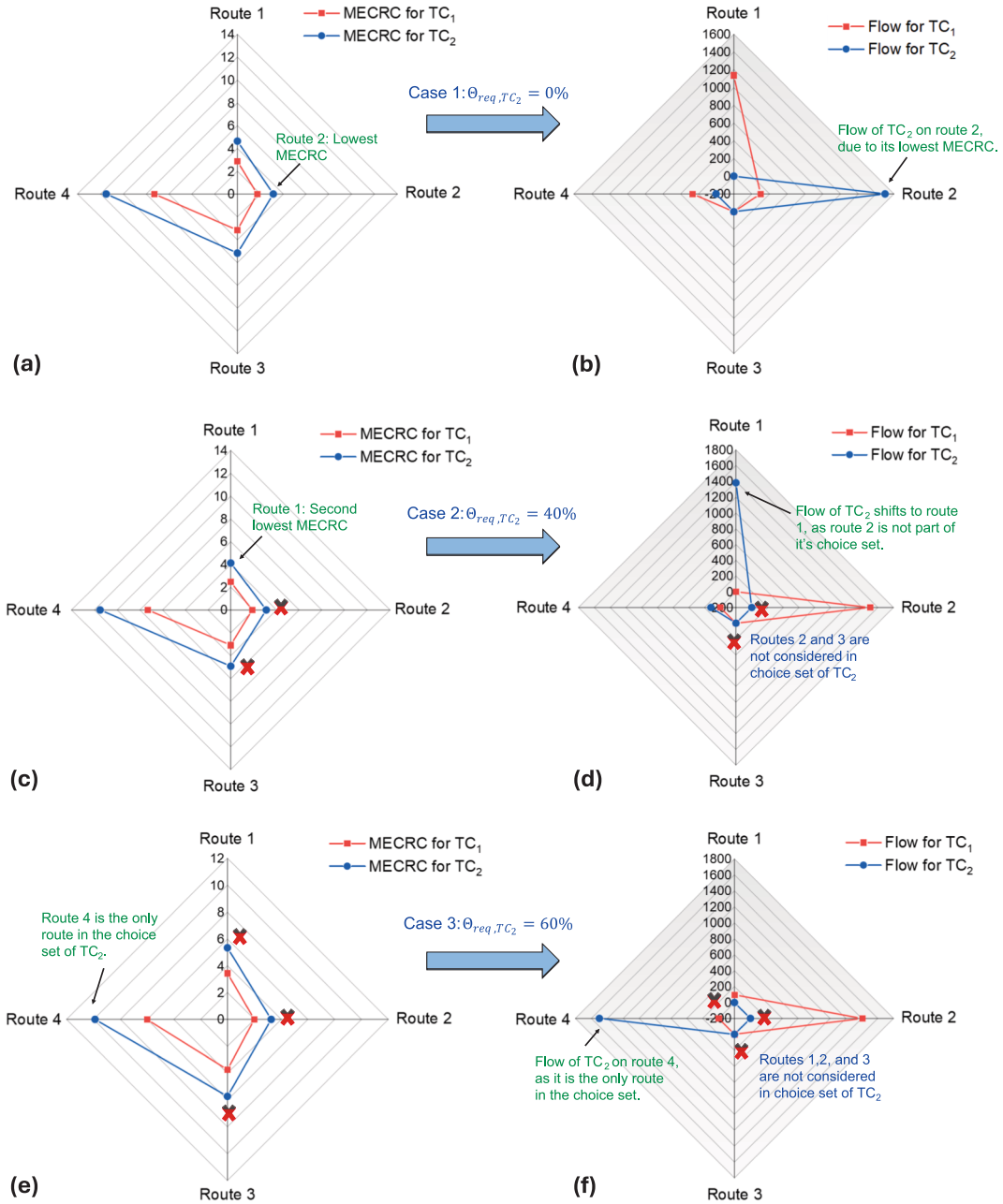


Fig. 10. Illustration of the results for the three cases. Subfigures (a,c, and e) present the MECRC for each case while Subfigures (b,d, and f) show the results for equilibrium route flows. In Subfigure (b), the flow of TC₂ is distributed on route 2 due to its lower MECRC. In Subfigures (d and f), as the route choice set of TC₂ becomes limited due to its higher safety score requirement ($\Theta_{req}^{TC_2}$), flow shifts to other routes even with higher MECRC.

are used by at least one of the two classes are shown for each OD pair.

For the OD pair (92,4), we see that the safety scores of both the routes (Fig. 14a) are higher than 50% thus both the routes are part of the choice set for TC₂. Route 2 has a higher safety score ($\Theta_2 = 68\%$) than route 1 ($\Theta_1 = 57\%$) because it has a lower share of local/collector roads compared to route 1. Route 2 has a local/collector road share of 27.2% compared to 60.7% for route 1. From Fig. 14b, we notice that TC₁ uses both routes 1 and 2, while TC₂, due to greater safety variability, opts only for route 2. Route 2 has a lower safety variability ($\sigma_2 = 4.44$), resulting in a lower MECRC.

For OD pair (79,103), both routes have safety scores above 50% (Fig. 15a), making them part of the choice set for both classes. However, as shown in Fig. 15b, TC₂ exclusively uses route 1 due to its lower CRC variability and MECRC, reflecting higher risk aversion, while TC₁ utilizes both routes. These results highlight the model's applicability to real-scale networks.

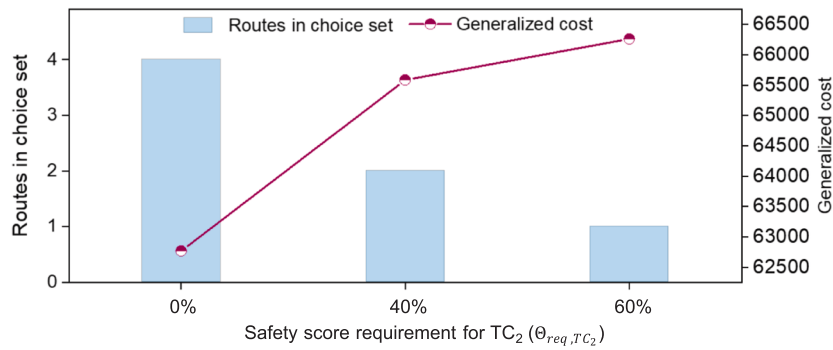


Fig. 11. Available routes in the choice set of TC₂ at different safety requirement levels and the corresponding generalized cost.

Table 10

Equilibrium flow of the multiclass travelers for case 4.

Class	Route	T_p	μ_p	σ_p	ψ_p	G_p	f_p	Used route
TC ₁	1	3.90	1.20	1.77	2.61	14.31	1298.69	✓
	2	4.23	0.65	1.20	1.61	14.31	201.31	✓
	3	5.33	1.50	1.88	3.00	19.00	0.00	×
	4	2.58	3.80	4.38	7.29	15.04	0.000	×
TC ₂	1	3.90	1.20	1.77	4.31	16.01	0.00	×
	2	4.23	0.65	1.20	2.77	15.47	1500.00	✓
	3	5.33	1.50	1.88	4.81	20.81	0.00	×
	4	2.58	3.80	4.38	11.49	19.24	0.00	×

Note: ✓ denotes used route, × denotes unused route.



Fig. 12. Winnipeg network showing the road functional class.

Fig. 16 illustrates the sensitivity analysis of route flows across different demand levels for each traveler class. From Fig. 16(a), it can be observed that as demand increases, the flow for the low safety-conscious class (TC₁) progressively shifts toward Route 2. In contrast, Fig. 16(b) shows that Route 2 is associated with a higher MECRC cost. Consequently, the high safety-conscious class (TC₂) allocates the majority of its flow to Route 1, with only a negligible proportion of users selecting Route 2. These results clearly highlight the influence



Fig. 13. Aggregate link flows for the traffic equilibrium model.

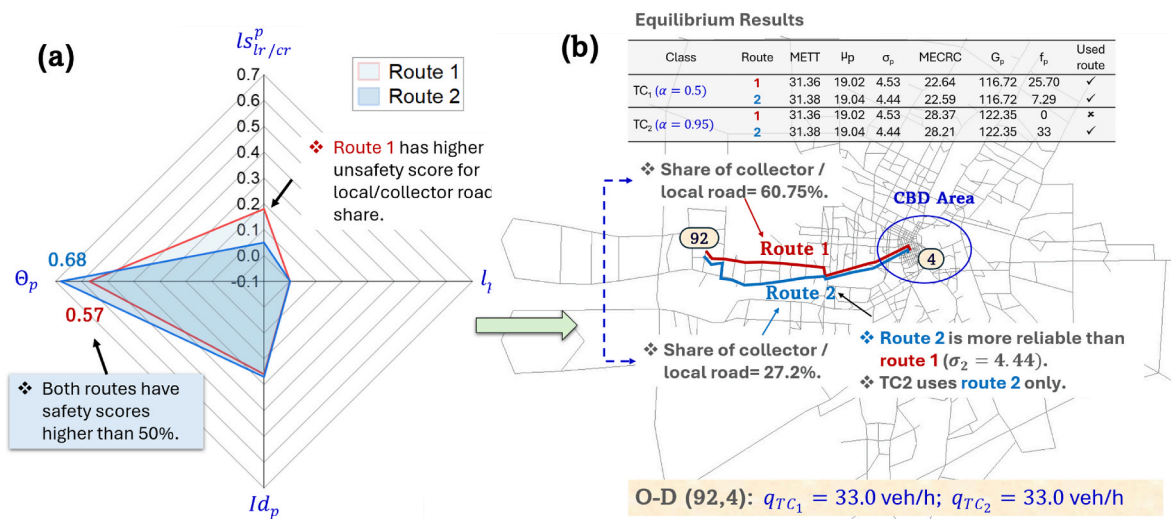


Fig. 14. Equilibrium results for multiclass travelers on OD pair (92,4) (a) Calculation of safety scores for the used routes (safety scores are expressed in %, e.g., 0.68 = 68%), (b) Equilibrium flow and travel costs on used routes.

of travelers' safety-consciousness on their route choice behavior under varying demand conditions.

5. Practical implications

By incorporating travelers' safety behavior in the network equilibrium model, the proposed traffic assignment model has significant potential for use by traffic planners, and policymakers for improving network safety. In the era of connected vehicles, route safety information will be disseminated to travelers, increasing its importance and demand in decision-making. The proposed model considers travelers' route choice behavior in light of crash risk as well as its consequences (i.e., the risk of fatal/severe crashes). It offers transportation planners a way to accommodate the effect of reshaping travel behavior on demand patterns due to travelers' safety concerns. Furthermore, this model provides an opportunity to systematically evaluate the effect of proposed traffic safety countermeasures and regulations on the traffic flow. Based on the estimation of travelers' responses to the safety regulation, the possible

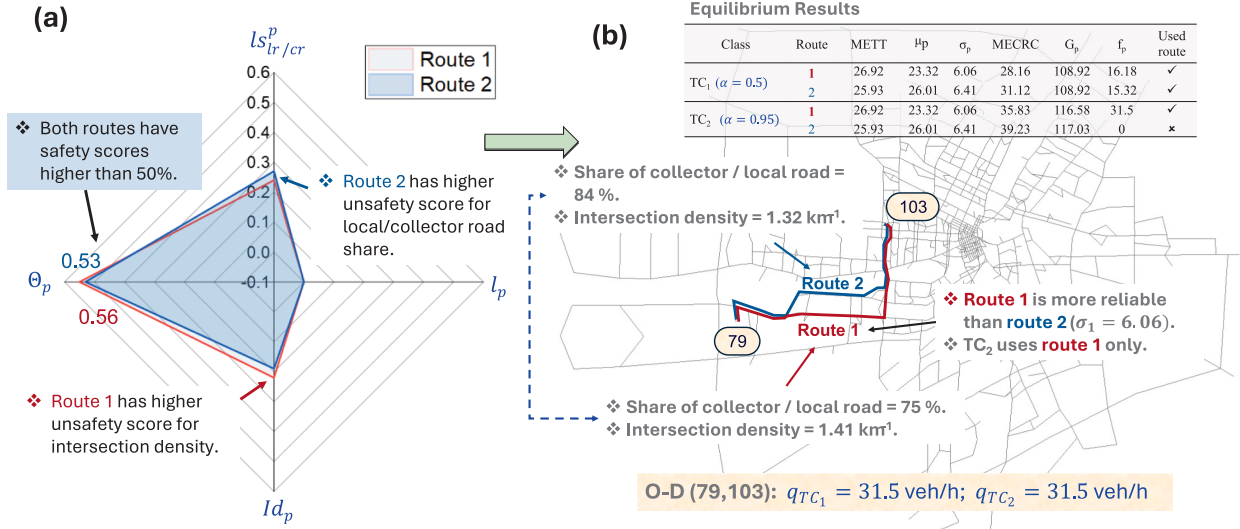


Fig. 15. Equilibrium results for multiclass travelers on OD pair (79,103) (a) Calculation of safety scores for the used routes (safety scores are expressed in %, e.g., 0.53 = 53%), (b) Equilibrium flow and travel costs on used routes.

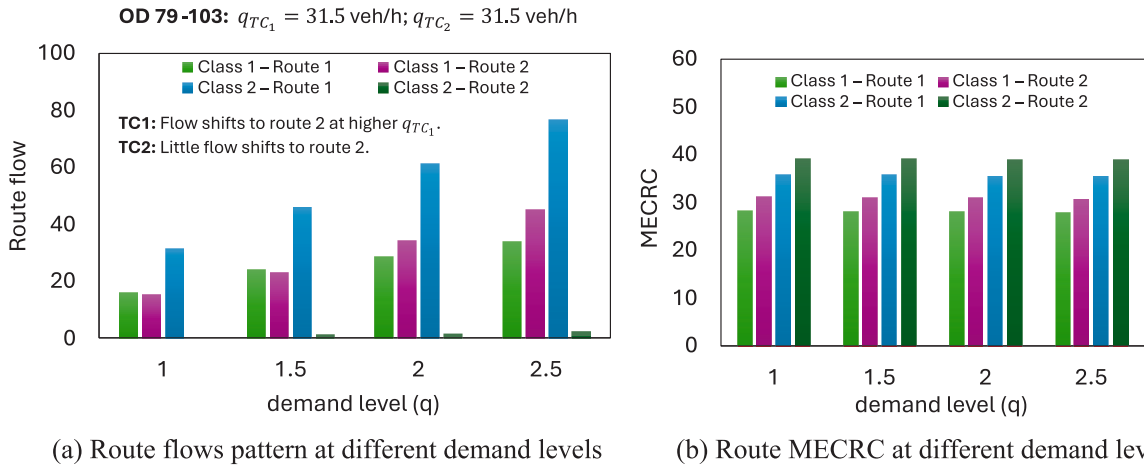


Fig. 16. Sensitivity analysis for route flows at different demand levels for OD pair 79 – 103.

changes in the flow pattern can be clearly identified. Thus, traffic planners and policymakers can develop safety improvement policies more systematically than by relying on historical data and experience.

Furthermore, the proposed model can be applied to design safety measures targeting the most vulnerable traveler class. Empirical research suggests that travelers with heterogeneous safety preferences exist on road networks (May et al., 2008). Gong and Fan (2017) found that reckless driving or over speeding significantly increases the chances of sustaining a fatal or severe injury. In a similar manner, distracted driving is found to negatively affect the driver's reaction towards an unsafe condition (Haque & Washington, 2015; Islam & Mannering, 2006). Thus, various countermeasures can be designed for routes accommodated by the vulnerable group of people to minimize the severity of crash effects. A number of countermeasures have proven effective in increasing road safety, including legislation and enforcement (Gitelman et al., 2012), safety awareness campaigns (Hoekstra & Wegman, 2011), and traffic calming measures (e.g., provision of speed humps). Safety education programs can be initiated through media and traffic police to target a specific group such as young drivers who are fond of reckless driving (Machin & Sankey, 2008).

6. Conclusions and future work

In this study, we propose a framework that integrates class-specific route choice sets while accounting for both the reliability and unreliability of the travelers' safety and travel time concerns within a network equilibrium model. The route choice set for each class is defined based on their safety score requirements, with the score of each route quantified by its roadway characteristics. The reliability

and unreliability aspects of safety variability are modeled using the α -reliable mean-excess concept. The traveler's safety variability concerns are modeled using a CRC distribution, and travelers are assumed to have prior knowledge of the distribution based on their long-term travel experience. For travelers with heterogeneous safety preferences, the proposed model ensures the reliability aspect of safe arrival more frequently at a predefined confidence level α_m and also accounts for the unreliability aspect of encountering the worst safety conditions beyond the effective CRC i.e., excessively large crash risk costs (unacceptable crash severity levels). A two-stage solution algorithm is employed to solve the non-additive route-based generalized cost, comprising METT and MECRC, in order to reach a user equilibrium solution. The proposed framework demonstrates that travelers with higher safety-score requirements have a more limited route choice set, while those with higher safety variability concerns tend to select routes with a lower CRC variability. The proposed framework shows how safety concerns influence the route choice and network equilibrium flow patterns.

Many future works can be explored to improve the proposed framework. First, additional factors related to roadway characteristics can be incorporated to quantify the safety score of a route. Second, from a modeling perspective, it would be valuable to model individual-specific risk averseness by adopting the continuous bicriteria traffic assignment framework, which can theoretically accommodate an infinite number of user classes (Xu et al., 2024). Third, empirical studies must be conducted to explore the travelers' safety preferences, specifically related to the reliability and unreliability of safety variability, and how travelers' trade-off between safety and travel time. Furthermore, the link CRC distribution, currently assumed to be normally distributed, serves as a conceptual representation of crash risk costs. However, validating the distribution parameters with empirical data is necessary to refine this assumption and calibrate it for different roadway types and traffic conditions. Finally, from a computational point of view, developing more efficient algorithms for solving the bi-objective non-additive generalized route cost would be beneficial.

CRedit authorship contribution statement

Umer Mansoor: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Zhengyang Li:** Writing – review & editing, Software. **Anthony Chen:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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